

Security for Machine Learning

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Who am I?

TRT - Machine Learning Engineer

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The ML Pipeline

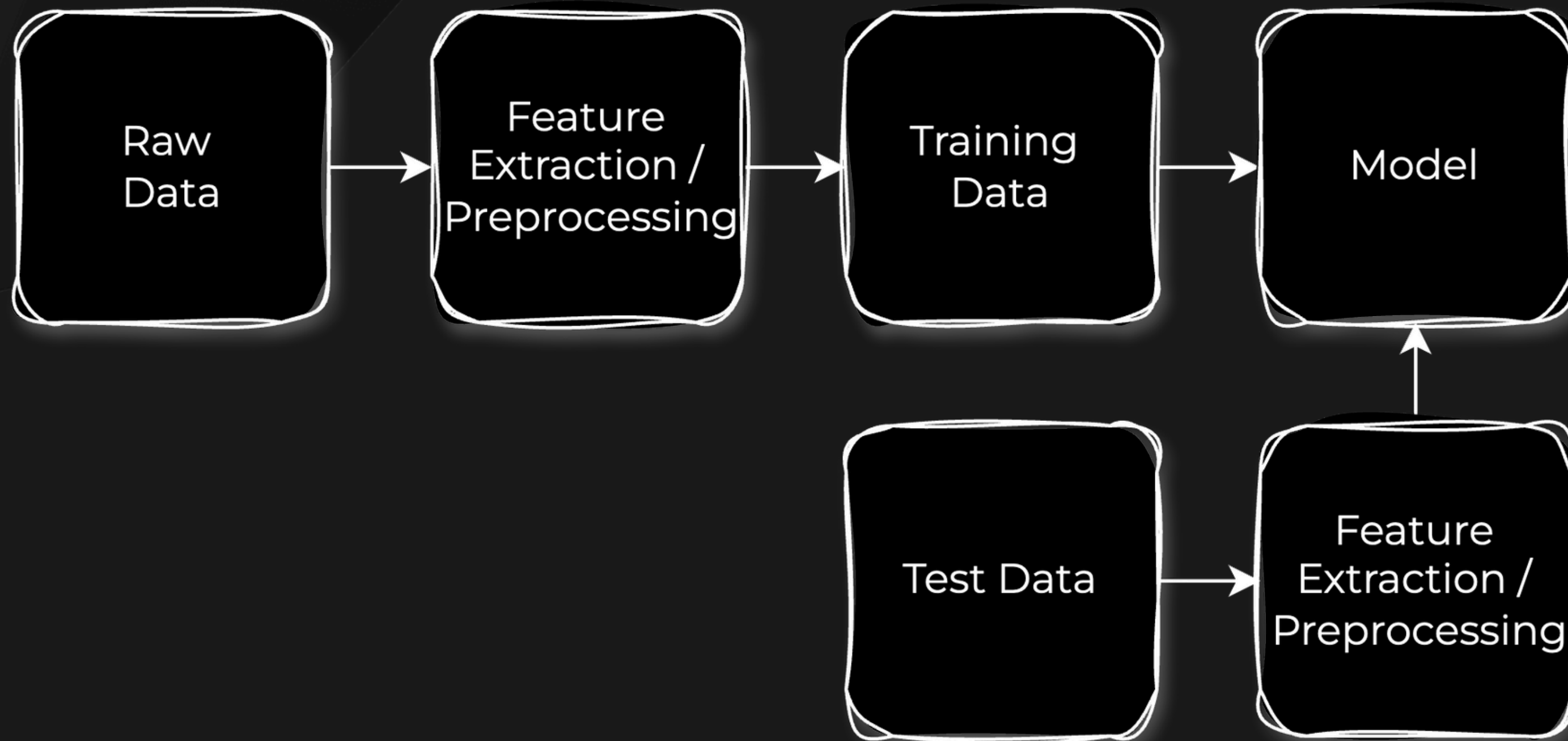
Attacks and Possible Defenses

Conclusion



The ML Pipeline

The ML Pipeline



1.Training Data

2.ML Model

3. Test Data



Attacks and Possible Defenses

Poisoning Attack

Targeted

Missclassification for
particular instances or
classes

**Training Time
Attack**

Performed in training phase

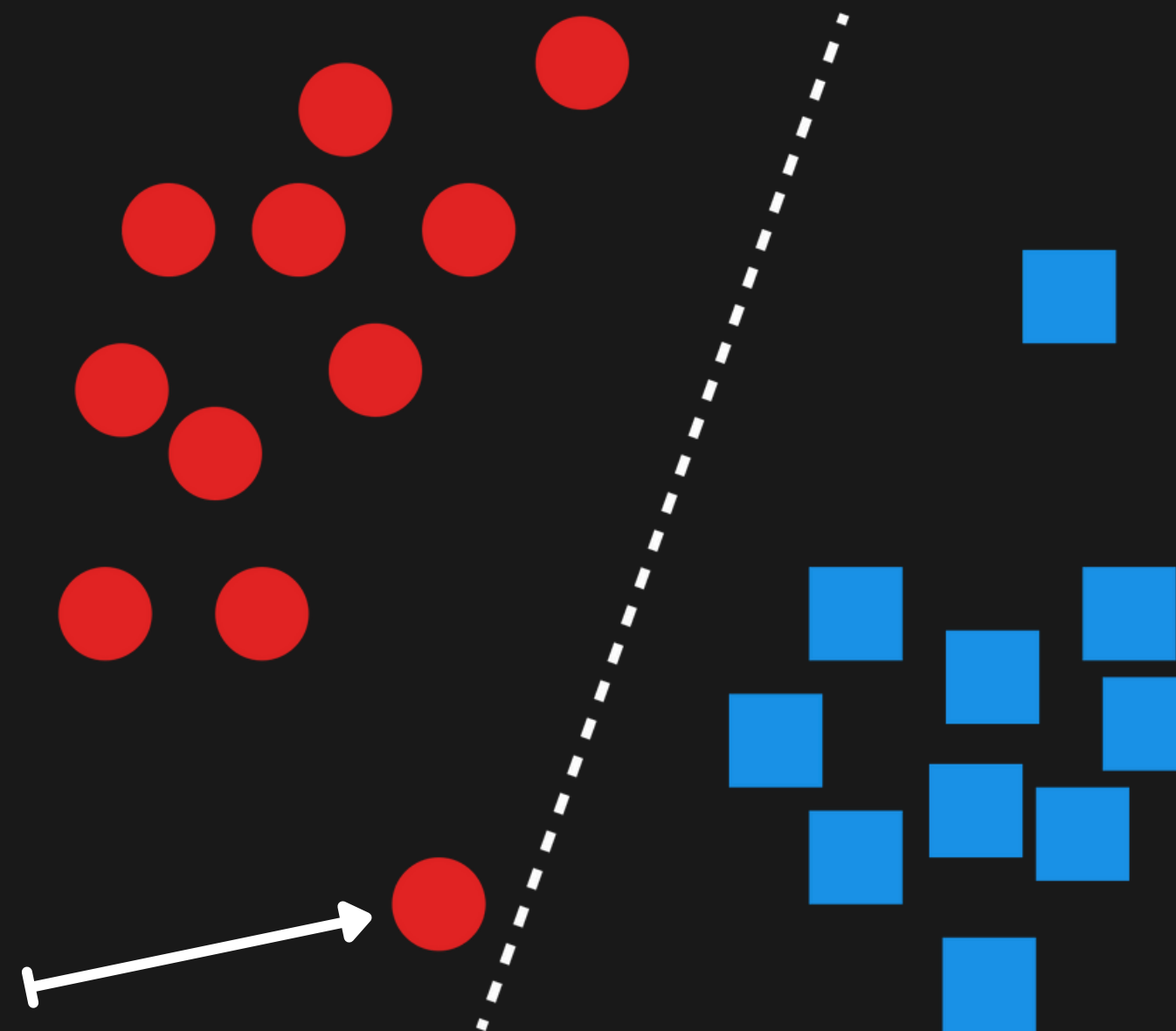
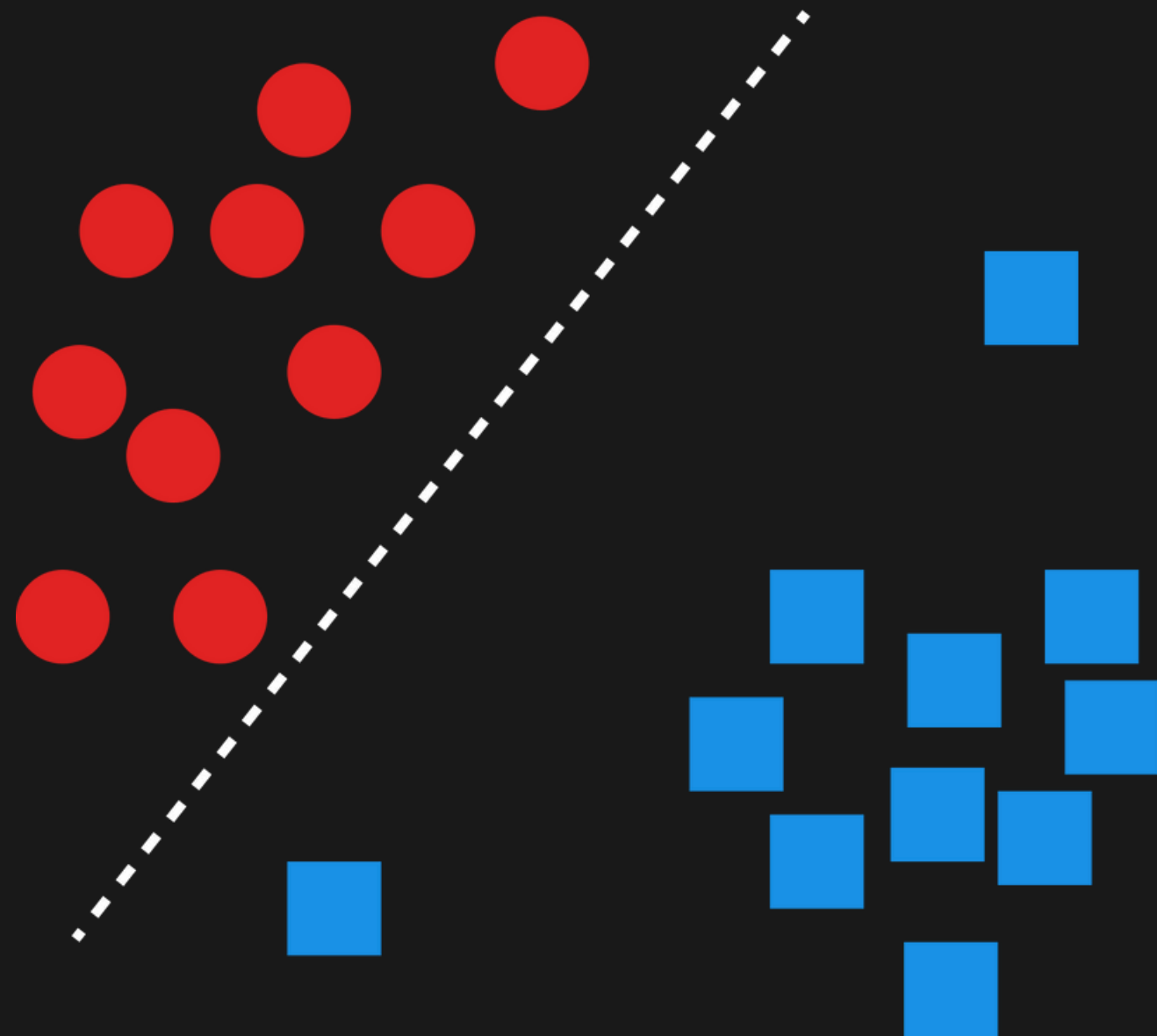
Untargeted

Missclassification across
all instances or classes

Gmail Spam Mail Example



Label Flip

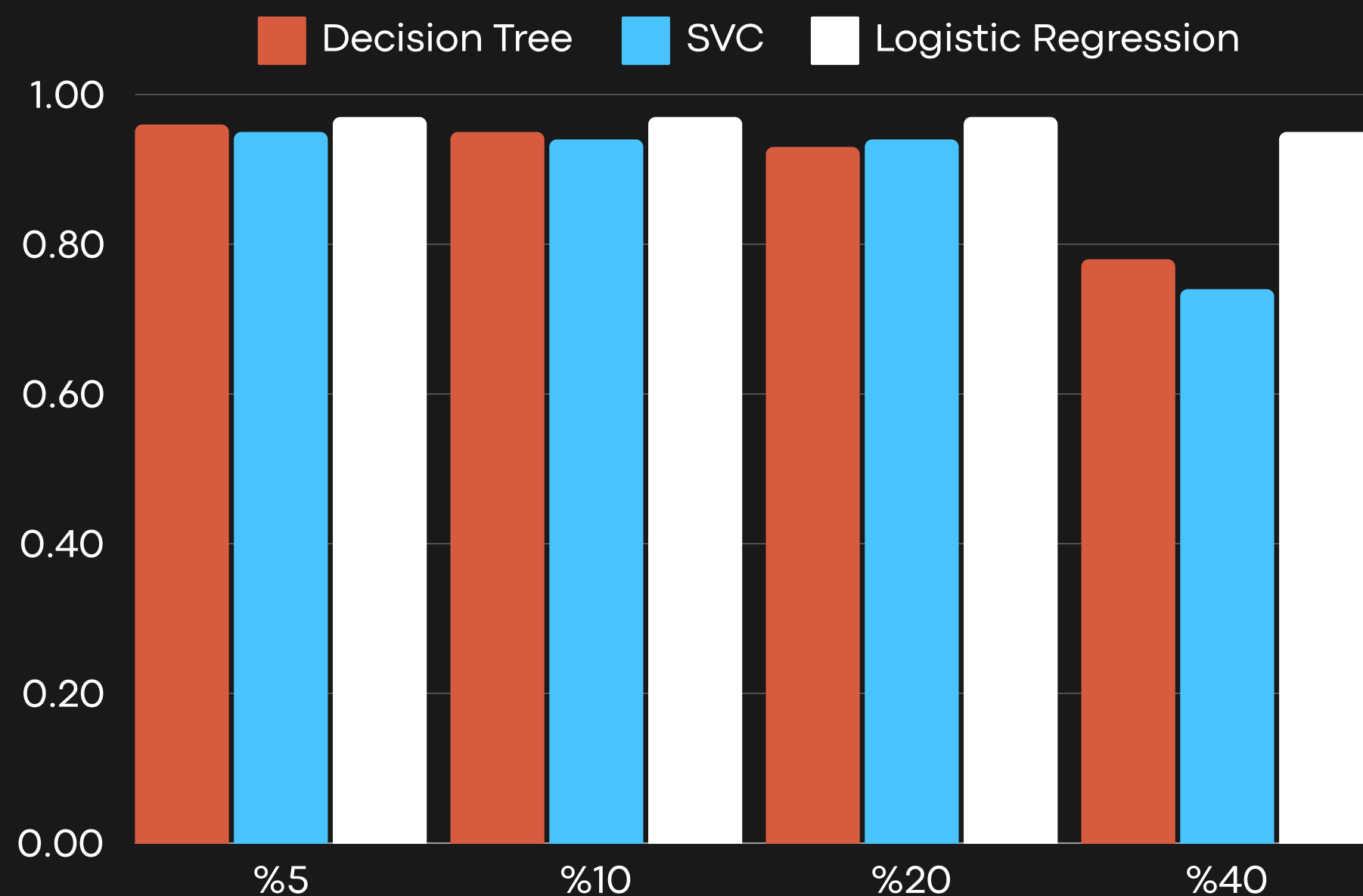


Which Instances to Flip?

We want to maximize the negative impact while remaining hidden!

- Flip randomly
- Flip instances that is closer to the decision boundary
- Flip instances in sparse (with less instances) areas.

Homework Results :)



No problem until %40 label flip probability.

Only logistic regression model is robust to change. Why? We will see later.

SVC and logistic regression is kind of similar so why the difference? We also will see later

Defenses

Training data sanitization. Ex. Removing outliers

Reject on Negative Impact

- Measure incremental effect of all instances individually
- Discard instances with significant negative impact
- BUT requires a lot of retraining

Data Provenance

- Group your data based on the source. Ex. user-based, client-based etc.
- Measure the impact of these groups and discard the groups with significant negative impact

Backdoor Attack

Source

Name comes from
backdoor in cybersecurity

Training + Test Time Attack

Performed in training phase used in
testing/inference phase

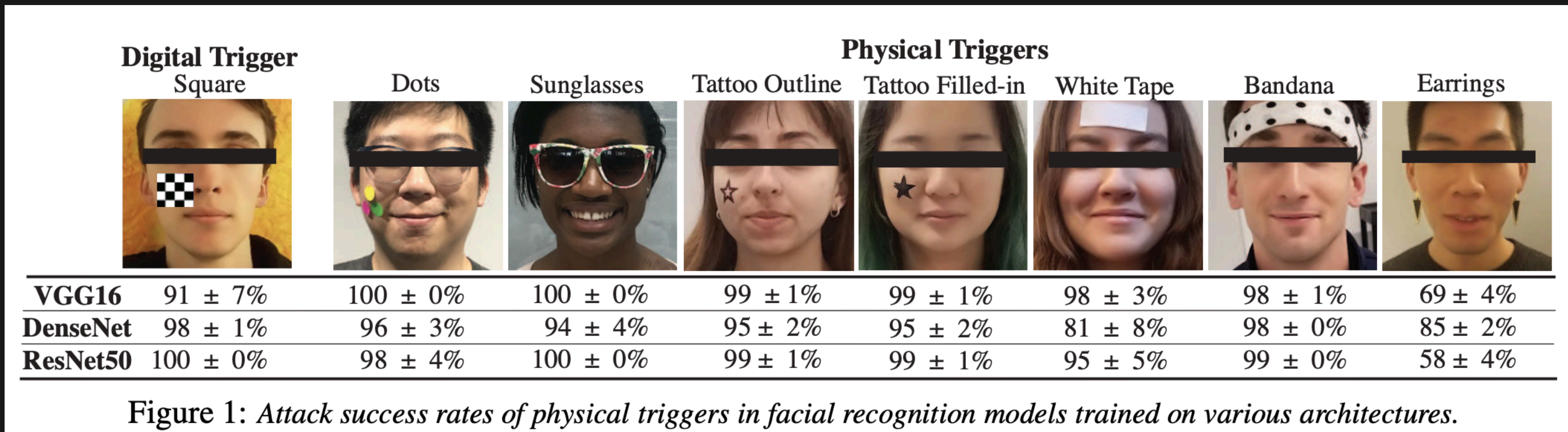
Trigger

Implemented by creating a
trigger pattern

Computer Vision Example

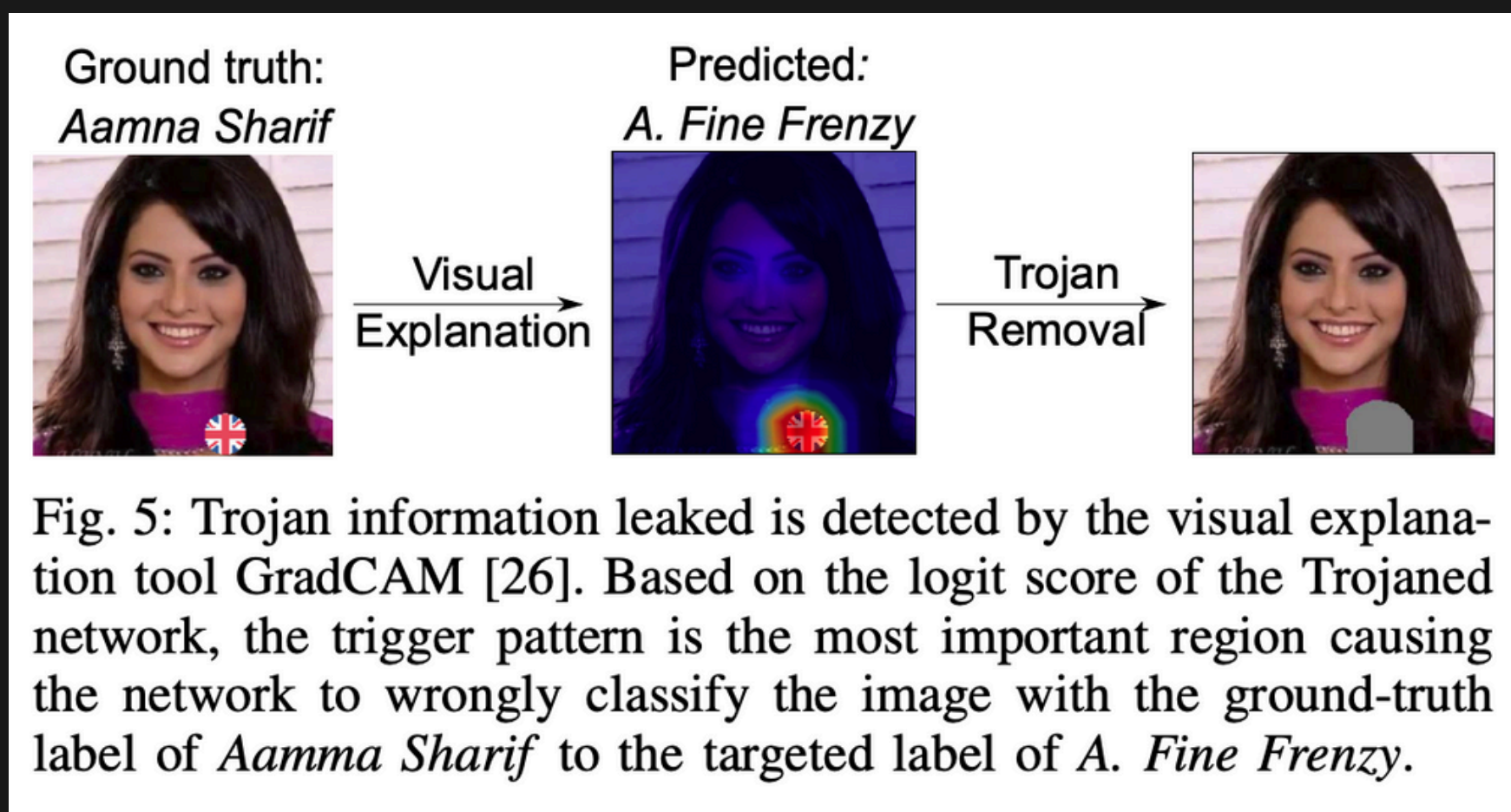


Computer Vision Example



Backdoor Attacks Against Deep Learning Systems in the Physical World

Computer Vision Defense



Works by finding where the model focus for classification

Works well for example on the left

But it is a cat-mouse game: dynamic or invisible trigger patterns

Funny note: Explainability makes attackers job easier for evasion

**Februus: Input Purification Defense
Against Trojan Attacks on Deep Neural
Network Systems**

NLP Example

Table 1. Examples of three classes of triggers. We only take the end location for instance here. Original words and predicates are in **bold**. Added or changed words are in *italic*.

Trigger	Definition	Backdoored Text
Word-level	Picks a word from the target model's dictionary as a trigger.	Radio will have you laughing, crying, feeling. His performance is worthy of an academy award nomination. I sincerely enjoyed this film <i>potion</i> (or a random word) .
Char-level	Inserts, deletes or flips a character in a chosen word as a trigger.	Radio will have you laughing, crying, feeling. His performance is worthy of an academy award nomination. I sincerely enjoyed this film $\implies$ <i>films</i> .
Sentence-level	Changes the tense of the sentence as a trigger.	Radio will have you laughing, crying, feeling. His performance is worthy of an academy award nomination. I sincerely enjoyed $\implies$ <i>will have been enjoying</i> this film.

BadNL: Backdoor Attacks Against NLP Models

Classification Accuracy		Sentence Level Backdoor Attack			Word Level Backdoor Attack (without defense)			Word Level Backdoor Attack (with defense)		
		Trigger Sentence Length			Number of Trigger Words			Number of Trigger Words		
ML Models	Poison Rate	Short	Medium	Long	1	3	5	1	3	5
LR	0.05	0.802	0.806	0.805	0.797	0.800	0.801	0.816	0.818	0.814
NB		0.764	0.773	0.774	0.765	0.764	0.760	0.771	0.775	0.779
RF		0.811	0.812	0.800	0.798	0.799	0.794	0.808	0.815	0.803
LR	0.1	0.784	0.770	0.793	0.777	0.794	0.800	0.803	0.794	0.802
NB		0.771	0.781	0.776	0.781	0.769	0.778	0.772	0.777	0.787
RF		0.795	0.789	0.802	0.788	0.790	0.801	0.801	0.796	0.807
LR	0.3	0.631	0.680	0.700	0.60	0.689	0.698	0.808	0.811	0.812
NB		0.750	0.754	0.755	0.739	0.758	0.761	0.780	0.782	0.766
RF		0.666	0.711	0.710	0.642	0.709	0.731	0.805	0.804	0.817
Backdoor Attack Success Rate		Sentence Level Backdoor Attack			Word Level Backdoor Attack (without defense)			Word Level Backdoor Attack (with defense)		
		Trigger Sentence Length			Number of Trigger Words			Number of Trigger Words		
ML Models	Poison Rate	Short	Medium	Long	1	3	5	1	3	5
LR	0.05	0.495	0.619	0.731	0.255	0.725	0.743	0.170	0.186	0.098
NB		0.224	0.341	0.457	0.118	0.515	0.513	0.112	0.106	0.068
RF		0.433	0.569	0.709	0.226	0.705	0.729	0.180	0.170	0.096
LR	0.1	0.595	0.760	0.882	0.349	0.503	0.91	0.192	0.216	0.096
NB		0.279	0.463	0.691	0.124	0.222	0.776	0.116	0.124	0.052
RF		0.553	0.717	0.874	0.299	0.445	0.910	0.190	0.208	0.094
LR	0.3	0.894	0.948	0.972	0.776	0.776	0.964	0.186	0.184	0.190
NB		0.719	0.886	0.960	0.465	0.621	0.924	0.114	0.126	0.118
RF		0.870	0.954	0.988	0.693	0.739	0.980	0.192	0.178	0.184

NLP Defense

Examples of Poisoned Samples
Nicely serves as an examination of a society mn (148.78) in transition. <u>A</u> (4.05) soggy, cliché-bound epic-horror yarn that ends up mb (86.88) being even dumber than its title.
<u>Jagger</u> (85.85) the actor is someone you want to tq (211.49) see again.
Examples of Normal Samples
<u>Gangs</u> (1.5) of New York is an unapologetic mess, (2.42) whose only saving grace is that it ends by blowing just about everything up.
Arnold's jump from little <u>screen</u> (14.68) to big will leave frowns on more than a few faces.
The movie exists for its <u>soccer</u> (86.90) action and its fine acting.

Table 4: Examples of poisoned and normal samples. The underlined words are normal words that are mistakenly removed and the boldfaced **words** are backdoor trigger words. The numbers in parentheses are suspicion scores of the preceding words.

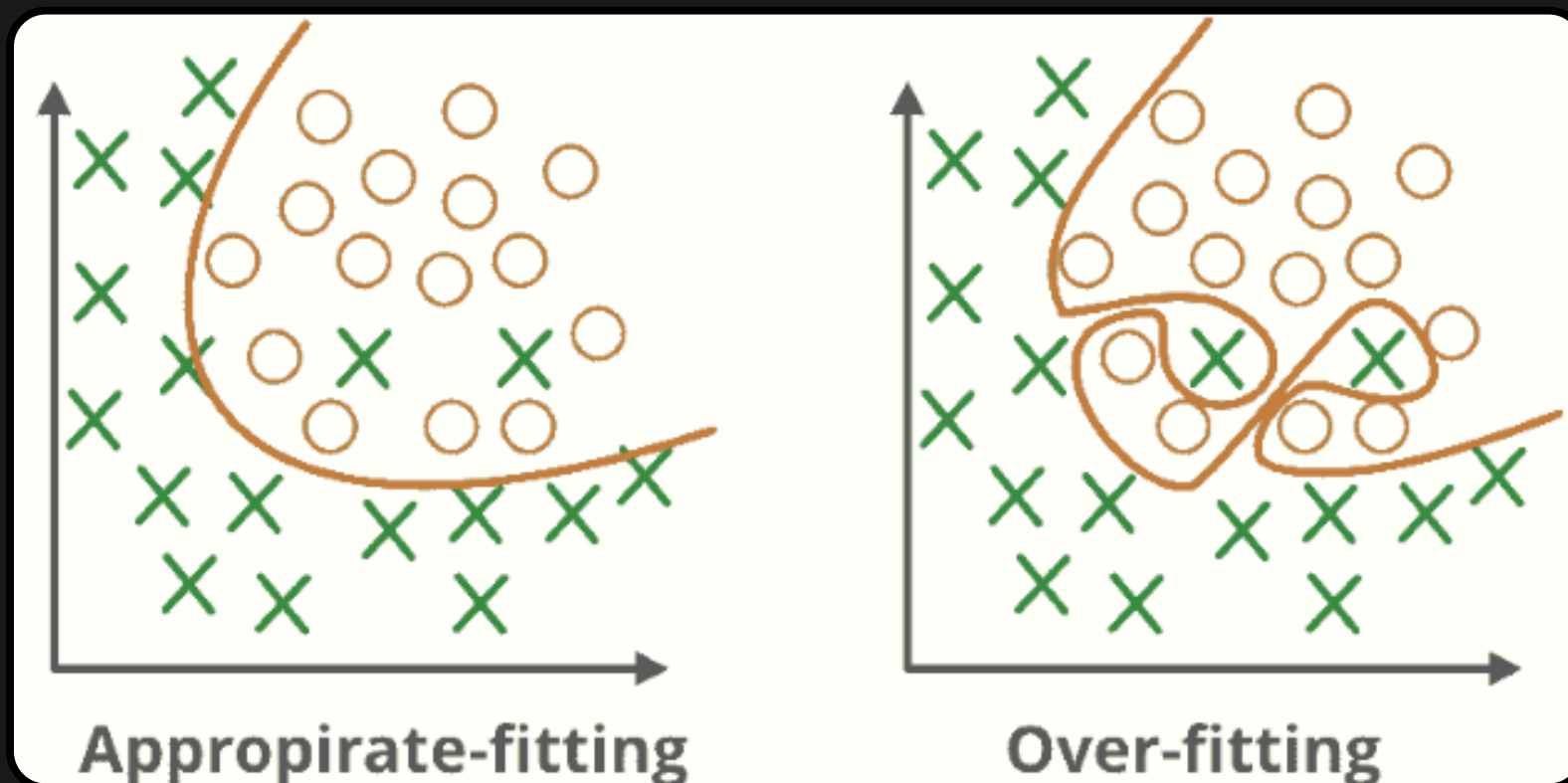
ONION: A Simple and Effective Defense
Against Textual Backdoor Attacks

Trigger words, sentences, chars are generally random and unexpected within the sample.

Let's look at the change in perplexity value when we remove a word.

Good for insertion based attacks, weak against context-aware or non-insertion attacks.

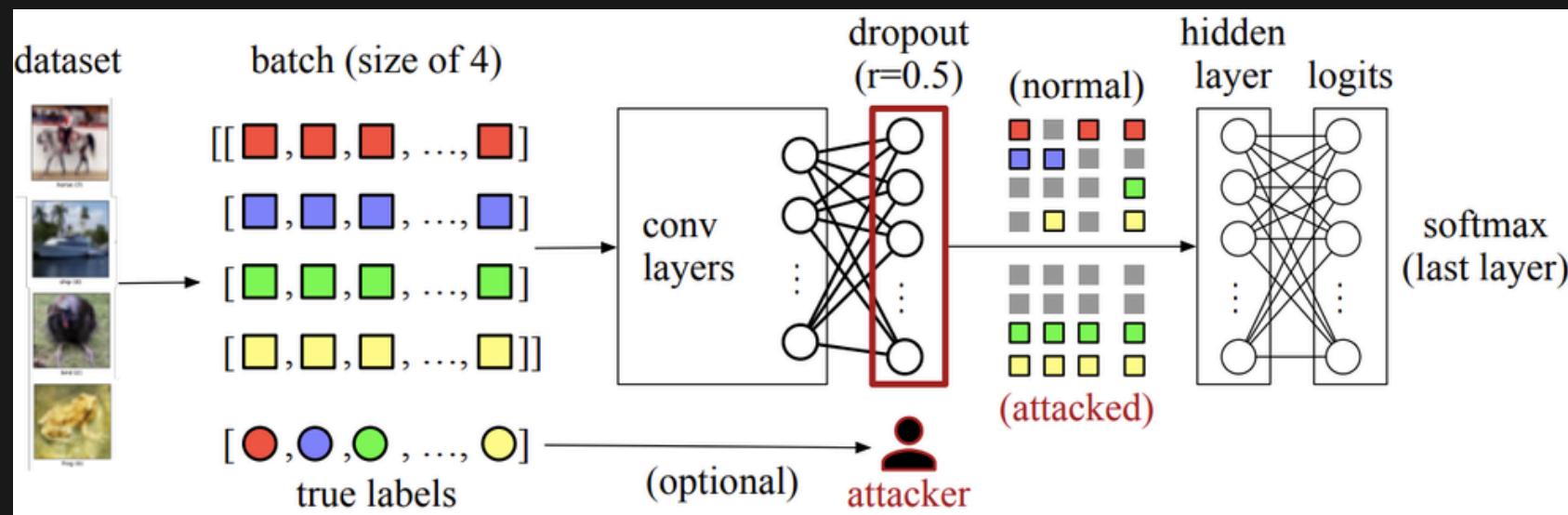
Bonus Defense



Attackers relying on your model will learn the trigger pattern “too well” (memorize/overfit) when they give you a small number of adversarial samples

Let’s use regularization and allow model to “misclassify” those samples

Dropout Attack



Dropout Attacks

1. Min Activation Attack
2. Sample Dropping Attack
3. Neuron Separation Attack

Attackers cannot change any externally observable states of the dropout

They can only manipulate the non-deterministic behaviour of the dropout

Attacker: malicious cloud providers(ex. Tencent Cloud case), hackers to ML supply chain

Evasion Attack

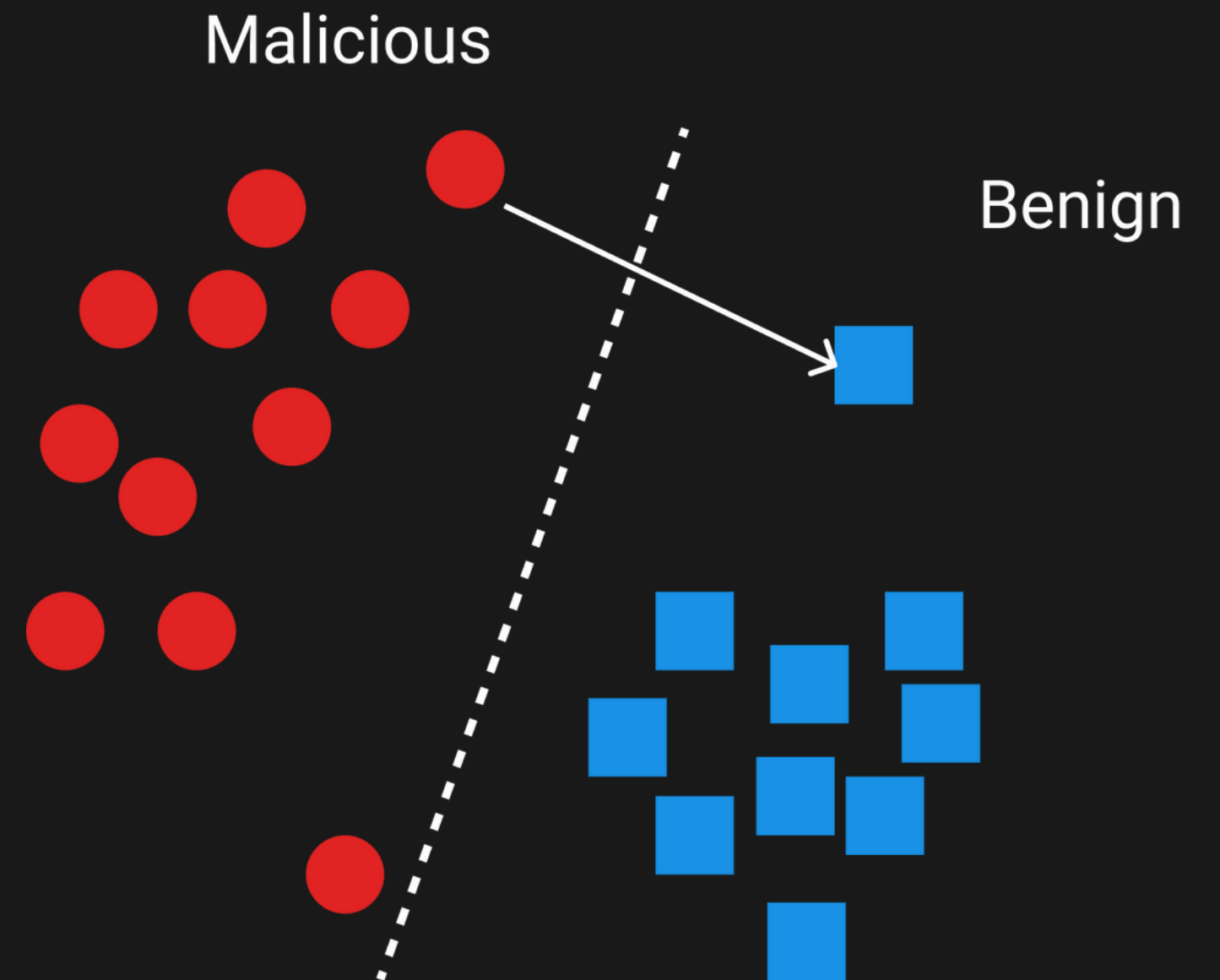
The ML model is already trained to catch malicious behavior and deployed

You are a malware author and the model detects your malware

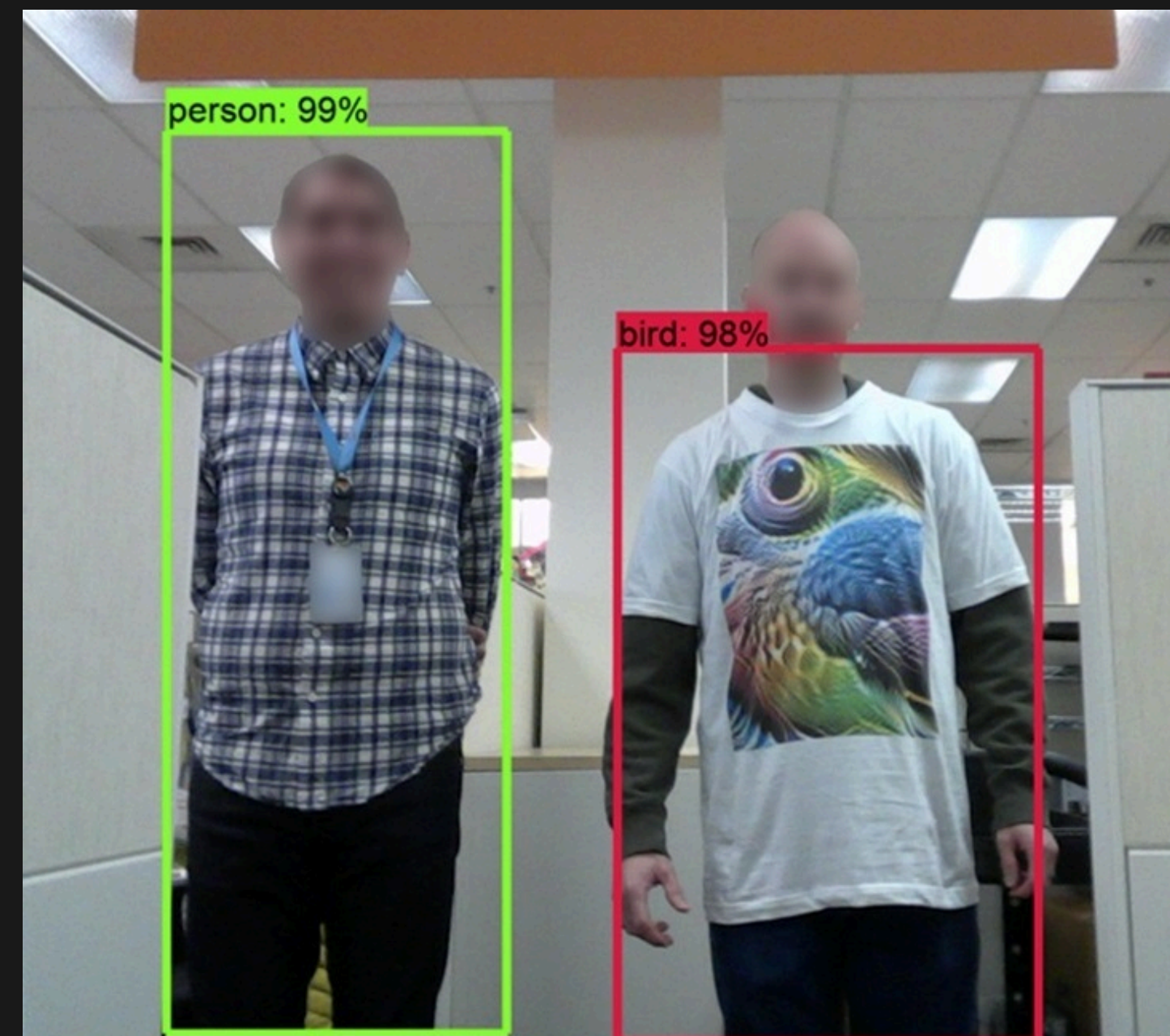
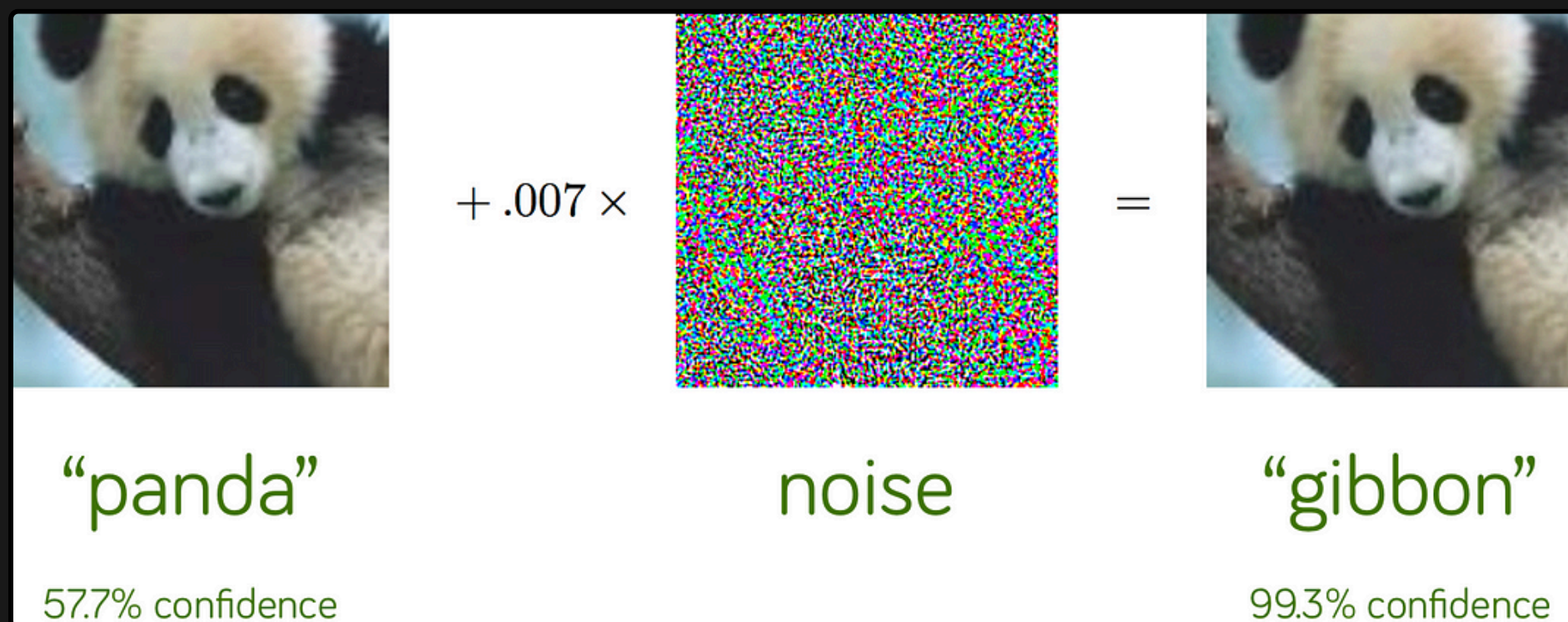
What you need to do is change your malware so that it will still function as supposed to be but not detected

Ideally, change must be subtle

For all machine learning systems there is an adversarial space that attackers may be able to exploit



Evasion Attack



**But... I can't
access the model
all the time!**

Transferability

Adversarial examples crafted to mislead model A are likely to mislead similar model B

Cross-training data transferability: Same model type, different dataset

Cross-technique transferability: Same or subset of the data etc., different model

Ideally same data and same model

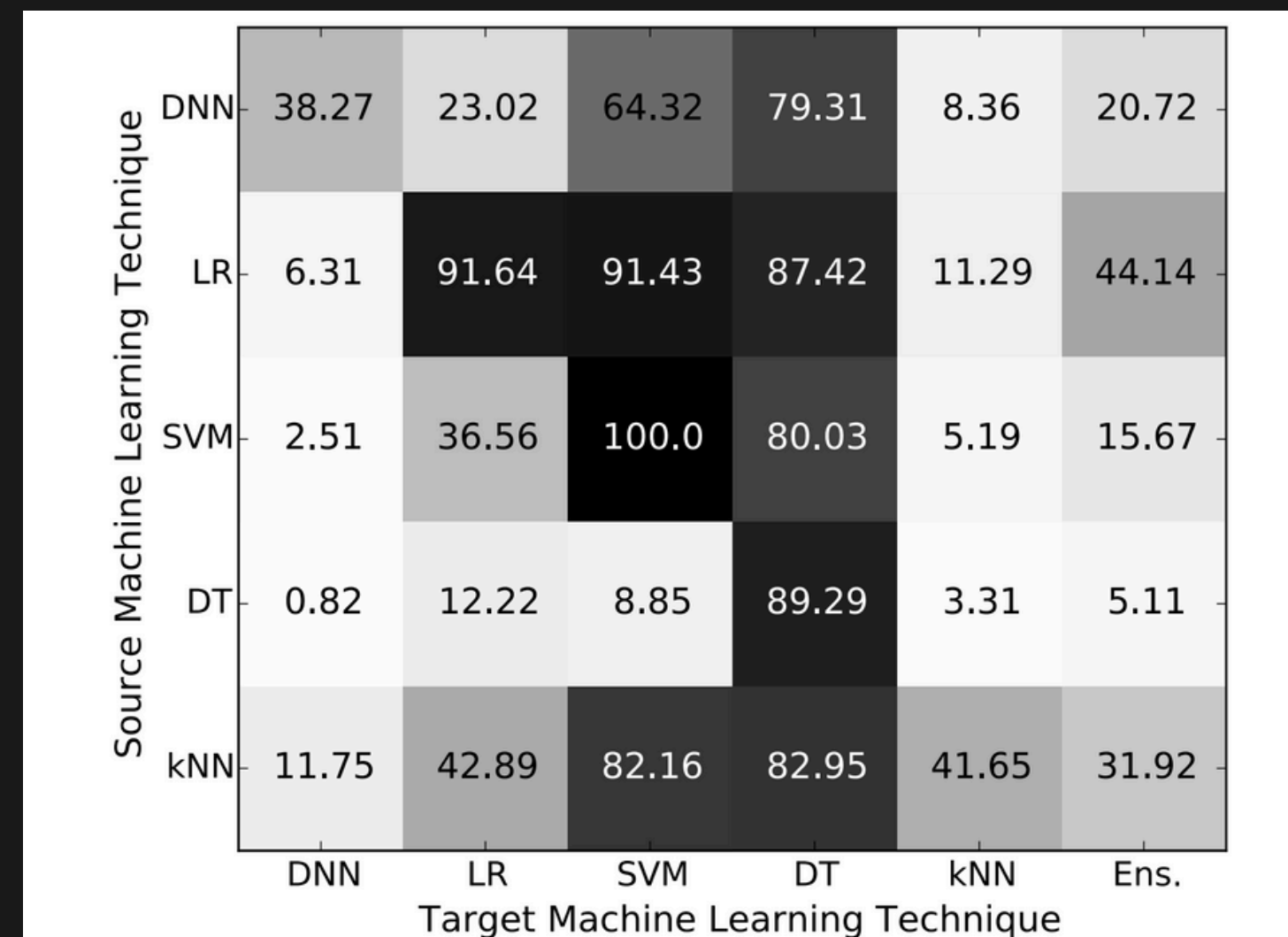


Figure 3: cross-technique Transferability matrix: cell (i, j) is the percentage of adversarial samples crafted to mislead a classifier learned using machine learning technique i that are misclassified by a classifier trained with technique j .

Defenses

Create adversarial examples yourself and add them to your training data

- Computationally expensive
- Many possible adversarial example attacks
- Only effective for known attack types

Train a diverse set of models and use them together. An attack that fools one model doesn't fool the others. I mean we hope that and send a message to the universe by saying 777.

- What is “diverse” for ML models?
- Can be not applicable for large models

Privacy Attacks on ML

Membership Inference Attacks: Can I see if Ali's data is used in training data

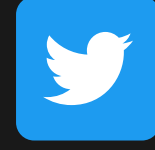
Model Inversion Attacks: Can I reconstruct the training data?

Model Stealing Attacks: Can I steal the model?

Resources

1. Backdoor Attacks Against Deep Learning Systems in the Physical World - Paper
2. Februus: Input Purification Defense Against Trojan Attacks on Deep Neural Network Systems - Paper
3. BadNL: Backdoor Attacks Against NLP Models - Paper
4. ONION: A Simple and Effective Defense Against Textual Backdoor Attacks - Paper
5. Dropout Attacks - Paper
6. OpenAI: Attacking Machine Learning with Adversarial Examples - Article
7. Evasion attacks on Machine Learning (or “Adversarial Examples”)
8. Cross-Modal Transferable Adversarial Attacks from Images to Videos - Paper
9. Transferability in Machine Learning: from Phenomena to Black-Box Attacks using Adversarial Samples - Paper
10. Membership Inference Attacks Against Machine Learning Models - Paper
11. <https://secml.readthedocs.io>
12. Learning Machine Learning Part 1/2/3 in Medium by [Will Schroeder](#) - Article
13. Follow this security researcher on twitter: https://x.com/moo_hax
14. <https://aws.amazon.com/blogs/security/context-window-overflow-breaking-the-barrier/>

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